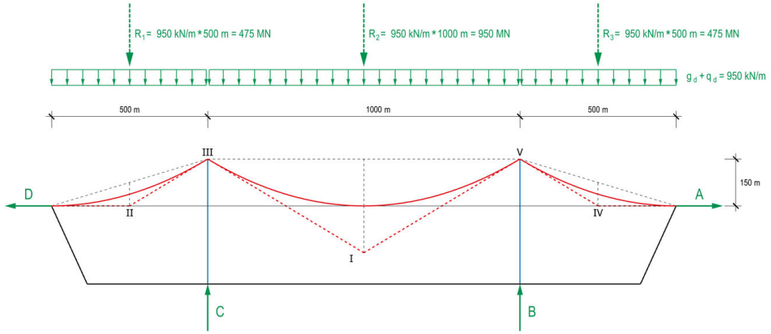
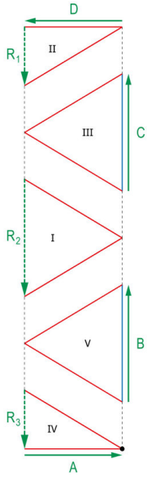
Cables

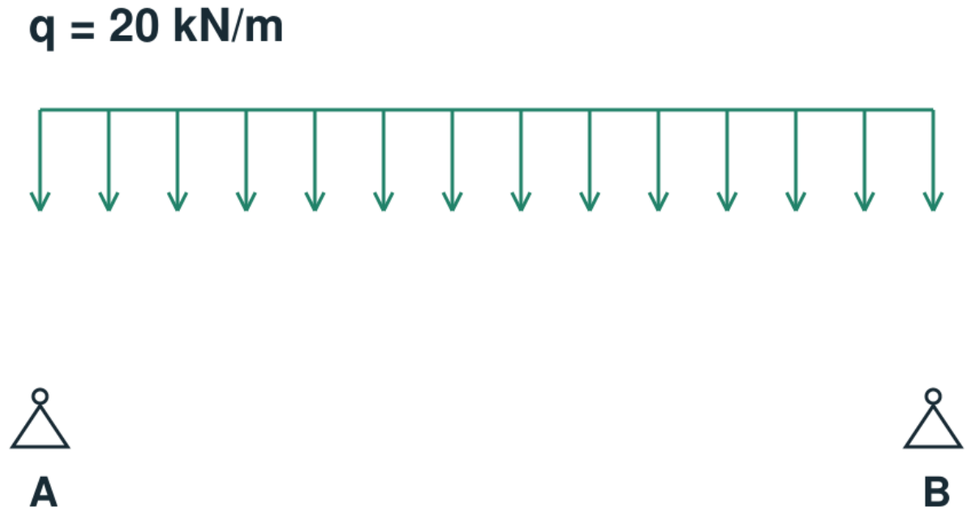


Golden Gate Bridge



Cable equilibrium





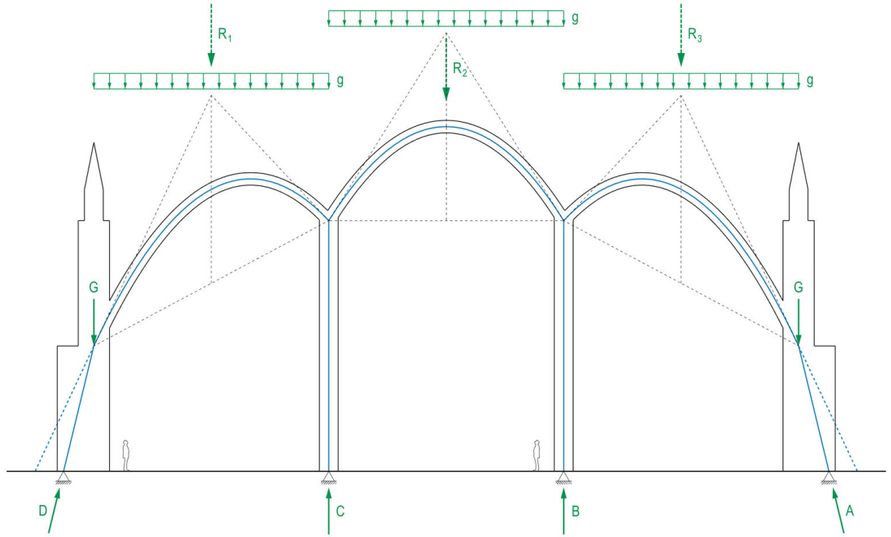
Cable form-finding

Draw a cable shape and its force diagram.

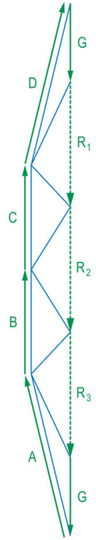
Arches

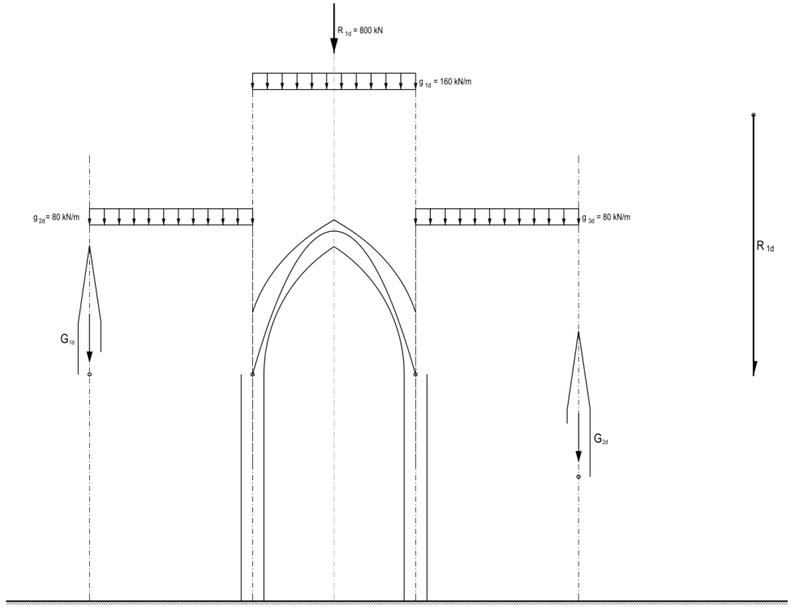


Beauvais Cathedral



Vault compression paths

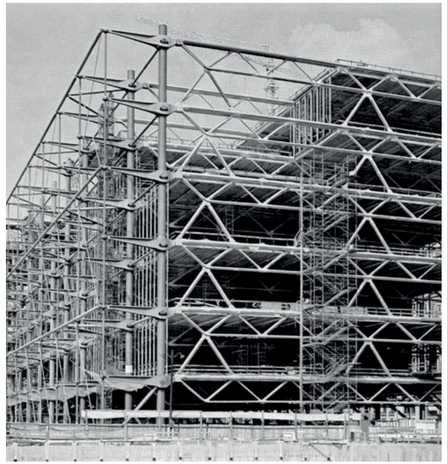




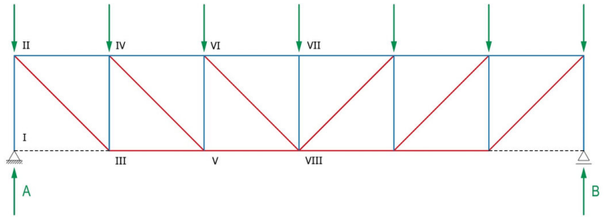
Side vaults and buttresses

Complete the side vaults and buttresses; draw the compression paths.

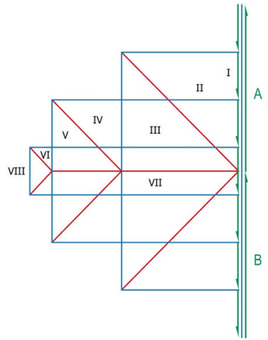
Trusses

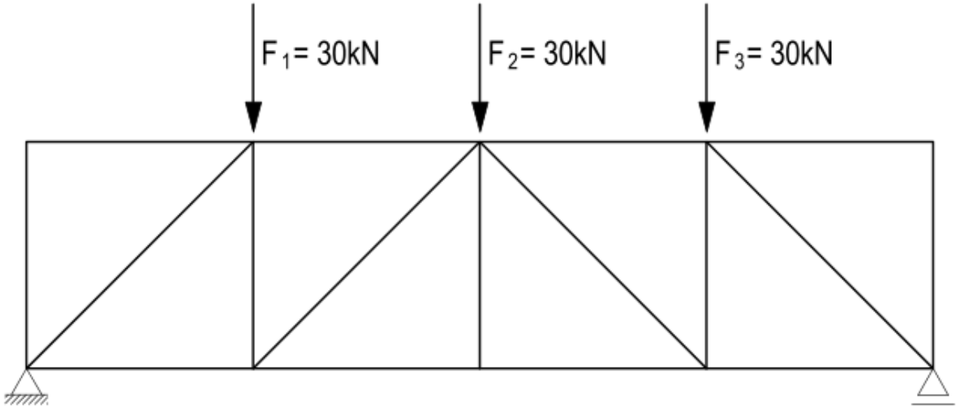


Centre Pompidou



Truss and force diagram





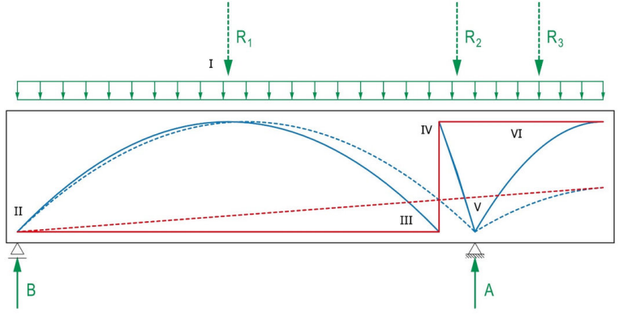
Truss member forces

Find support reactions and construct the member forces.

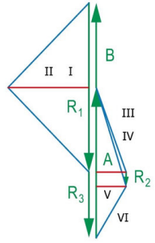
Beams

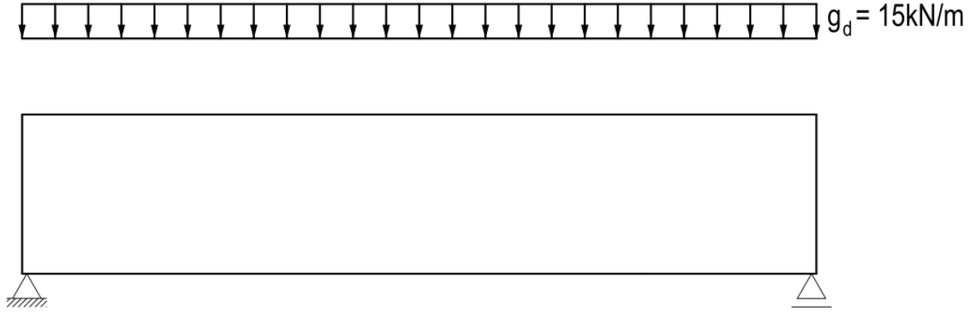


House Kronleinstrasse



Beam force flow





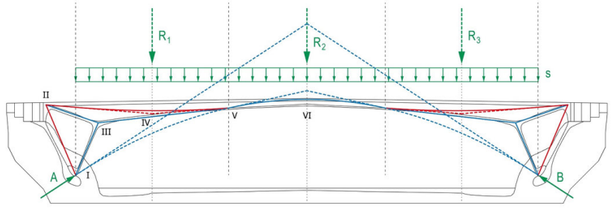
Simply supported beam

Draw the internal force flow and find support reactions.

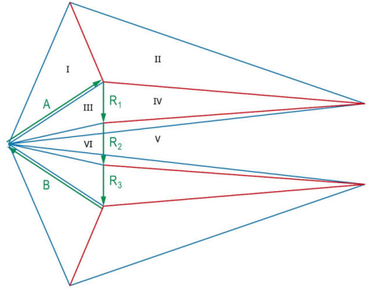
Frames

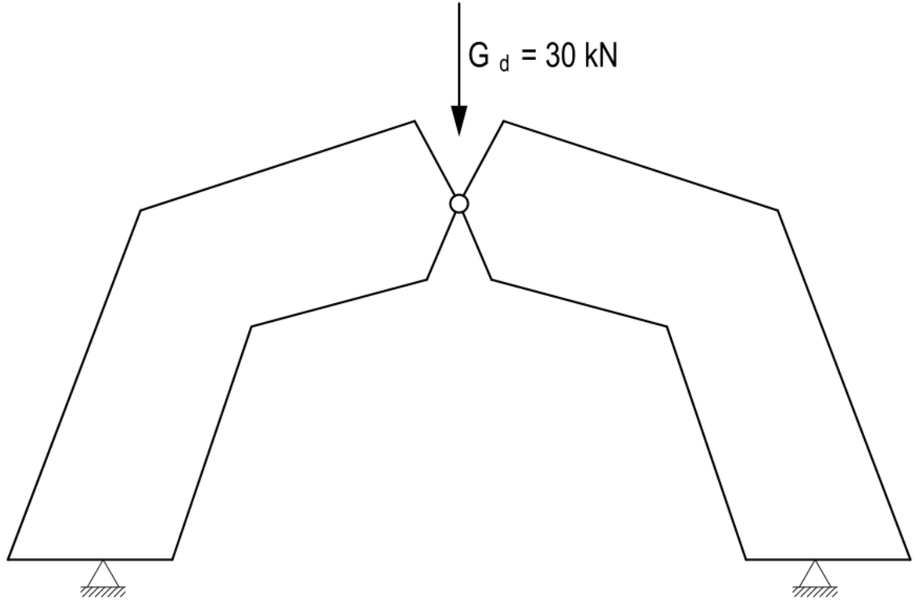


Tagliamento Bridge



Frame equilibrium

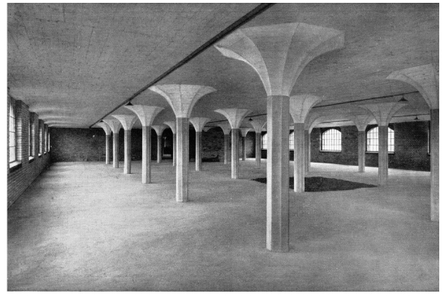




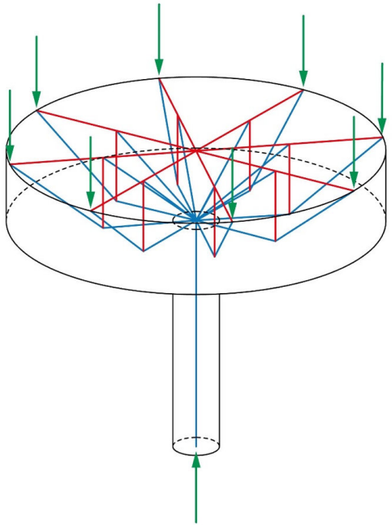
Three-hinged frame

Construct a force path through the frame and close the force diagram.

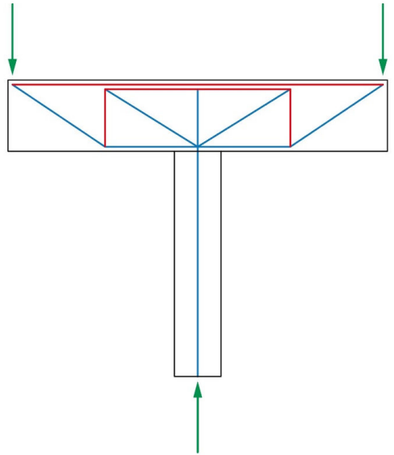
Plates and slabs



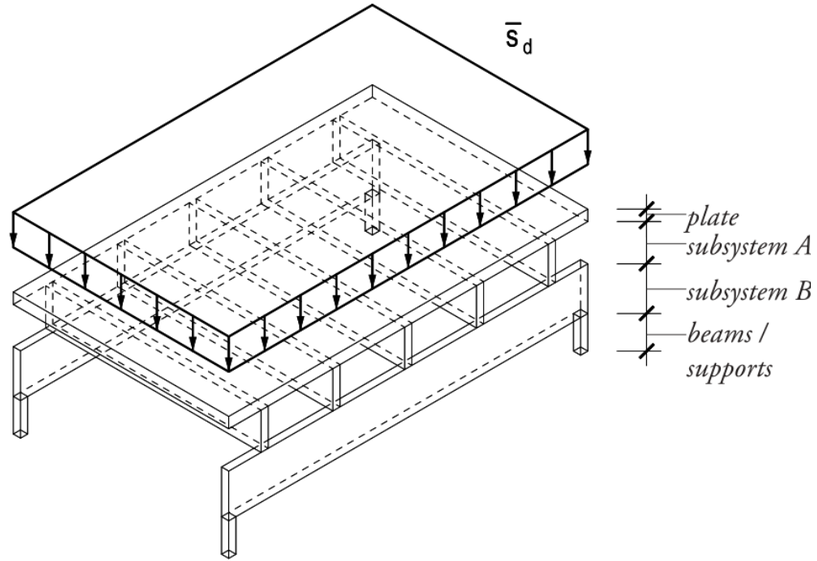
Mushroom slab



Slab force flow



a)



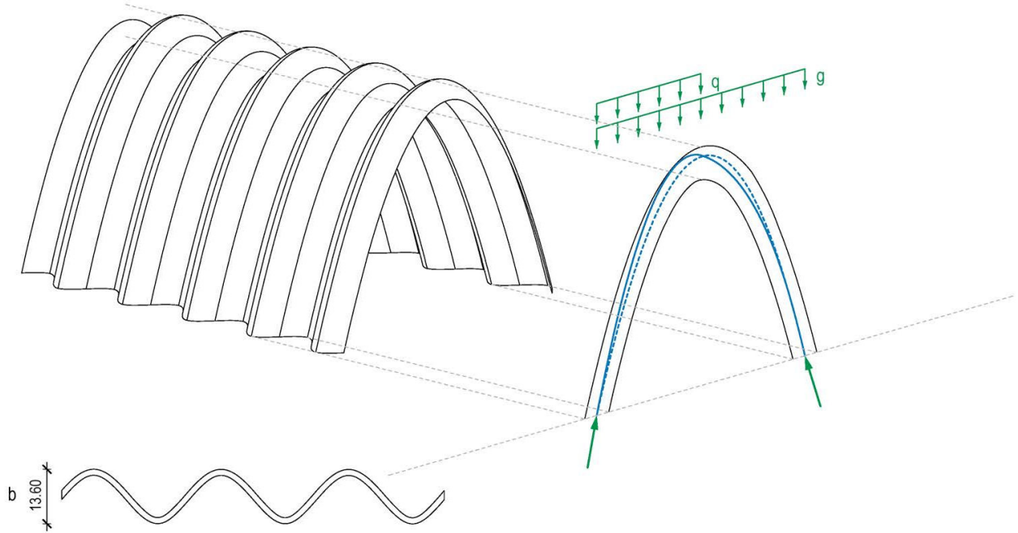
Slab and supporting beams

Trace the slab loads through the beams to the supports.

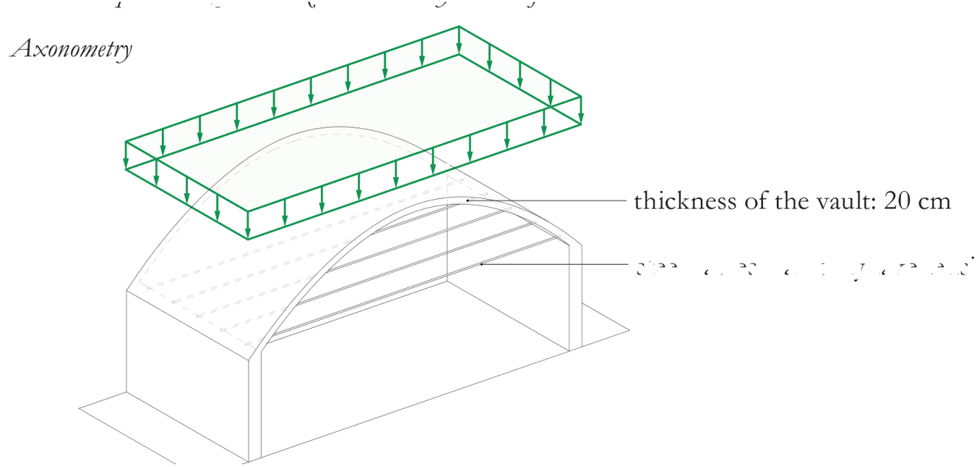
Shells and vaults



Cadyl Silo



Shell as arch strips



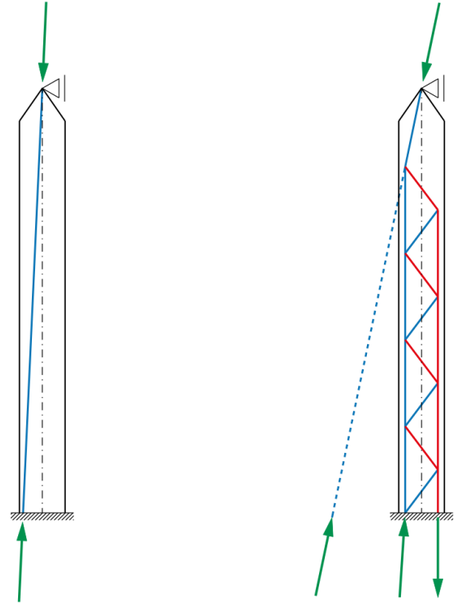
Tied masonry vault

Draw the vault compression paths and find the tie forces.

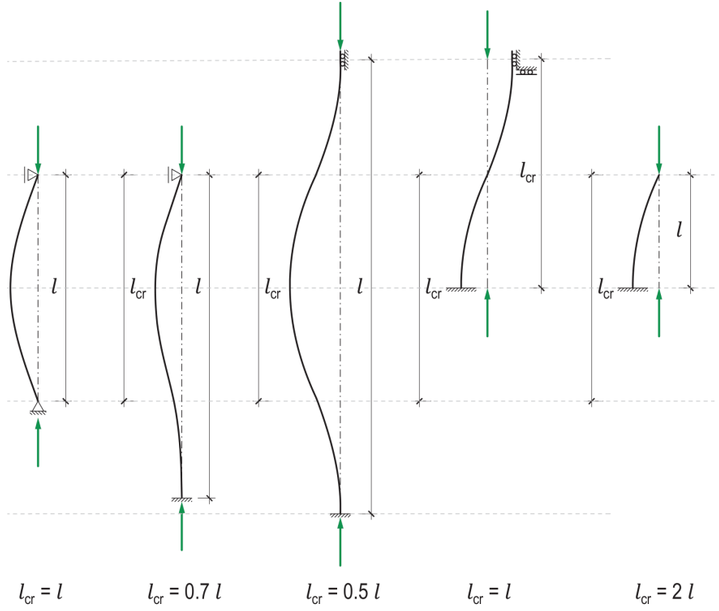
Columns



Carre d’Art



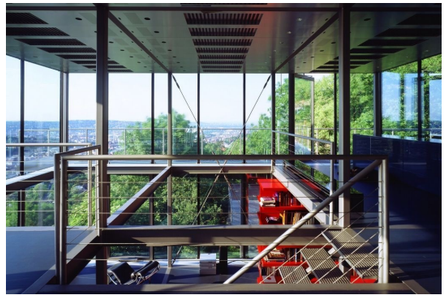
Column force flow



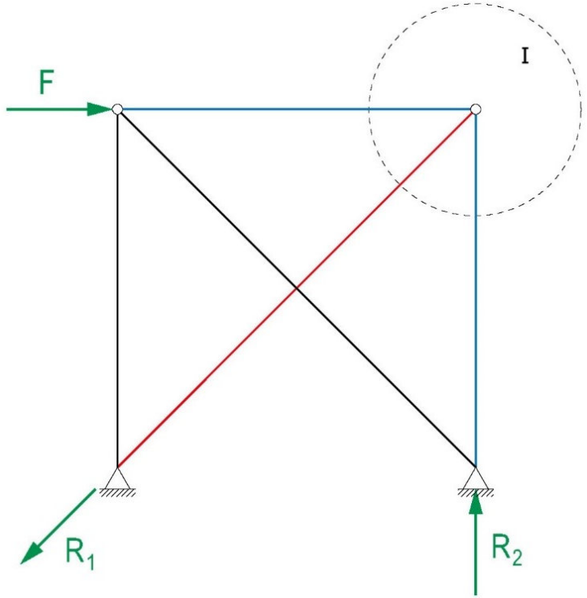
Column support conditions

Determine the buckling length for each support condition.

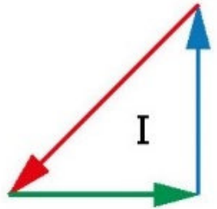
Bracing

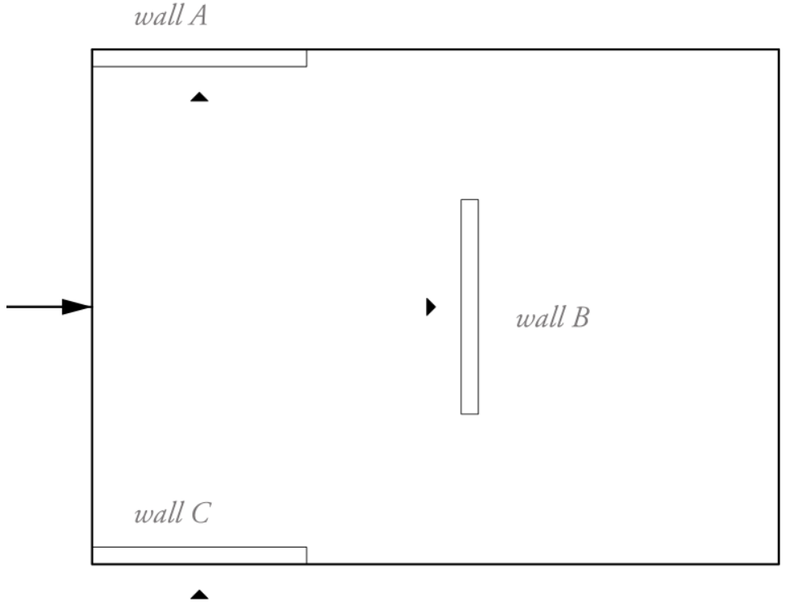


House R128



Bracing equilibrium





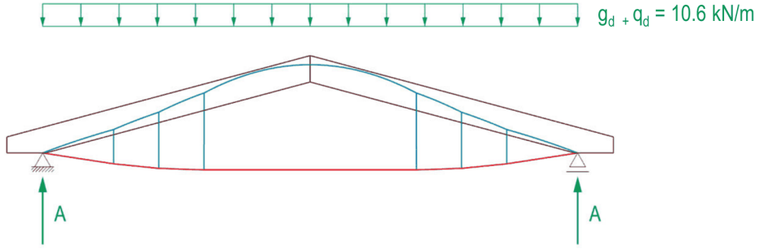
Horizontal load transfer

Transfer the horizontal load through the walls to the supports.

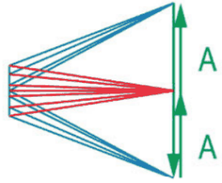
Material-specific forms

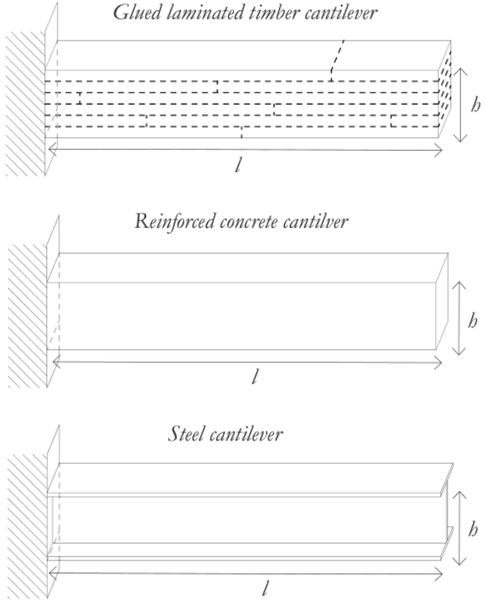


Gym Vrin



Timber roof equilibrium





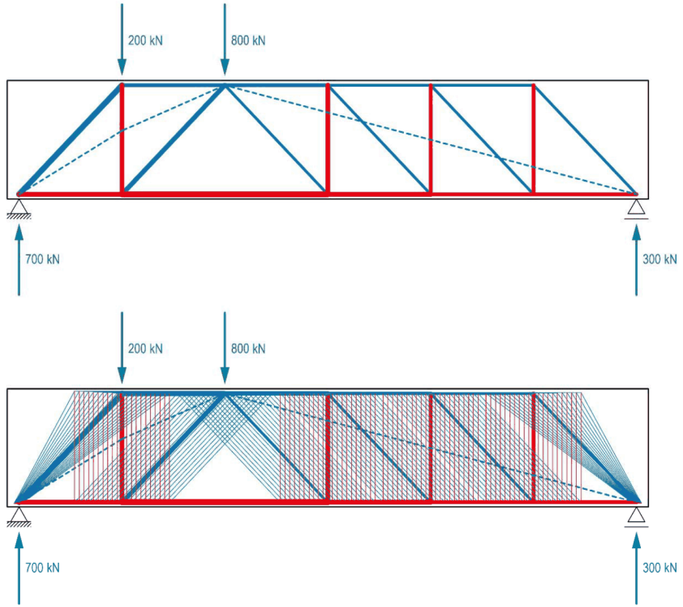
Cantilevers in three materials

Compare the required cantilever depth in timber, concrete and steel.

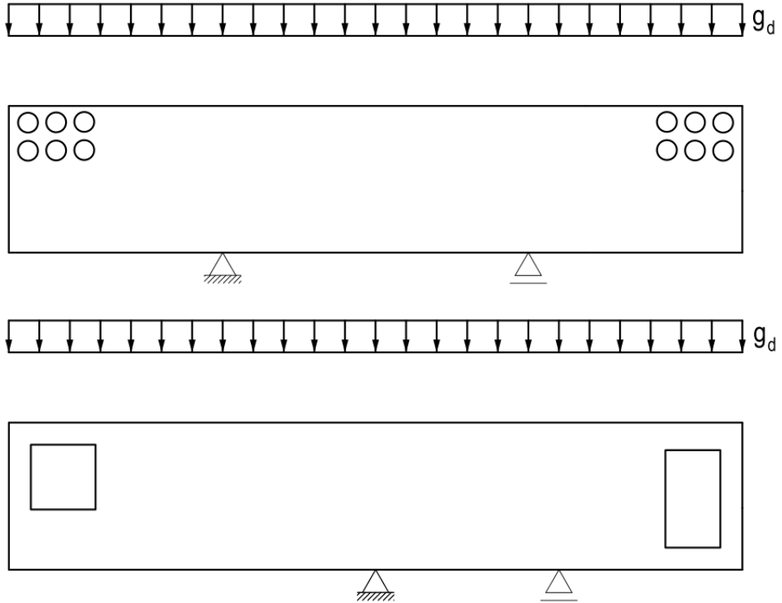
Plastic and stress-field models



Palestra Doppia



Strut-and-tie and stress field



Beams with openings

Draw strut-and-tie force paths around the openings.